

CS772

Deep Learning for Natural Language Processing

Midsem

01/02/24

Time- 4.15am to 5.00 pm

marks-30

(All questions are multiple choice questions, except Q9 and Q10. For Q1 to Q8, more than one answer may be possible, in which case all of them have to be given. If the answer is wrong which may include not choosing all the correct options, you get -1; else 2 marks for each. Q9 is of 9 marks and Q10 of 5 marks. Use the supplied A4 sheet for answering Q0 and Q10. No extra sheet will be given.)

Q1. Recall the word vector finding network discussed in the class with the words ‘heavy’, ‘light’, ‘rain’ and ‘shower’. The network has 4 input neurons U_0, U_1, U_2, U_3 , 2 hidden neurons H_0, H_1 , and 4 output neurons V_0, V_1, V_2, V_3 . The initialized weights are as follows:

Into H_0 : all weights are 1; into H_1 : all weights are 1.5

Out of H_0 : all weights are -1; Out of H_1 : all weights are -1.5;

Identify the statement(s) that is/are DEFINITELY TRUE:

- (a) The network with the given initialization will converge giving appropriate word vectors, given enough data, resource etc.
- (b) The network will not converge and serve its purpose, unless symmetry breaking is done
- (c) For these set of weights, the momentum factor should not be used
- (d) None of the other options is true

Ans: (a)

Justification: There is no problem with the n/w. (a) surely holds. Seemingly there is a symmetry breaking problem. But that is not true. Weights radiating from input neurons should not have symmetry about the vertical axis, as also weights entering into the output neurons.

Q2. The probabilistic BP (PBP) algorithm goes as follows. A weight in a network (which has many weights) is changed by gradient descent rule with probability P , and by an amount $c \cdot |w|$ with probability $(1-P)$, where c is a fraction and $|w|$ is magnitude of w . For a particular instance of PBP, a weight w is such that the loss L vs. w curve (a projection in the L - w plane) has a local minimum at $\langle w=0, L=0 \rangle$ and $L=0.5w^2$ in a small neighbourhood of this minimum. For a weight value of $w=-0.01$ and learning rate of 0.1, which of the conditions on c can be expected to make the n/w bypass the local minimum in a single step at this value of w ?

- (a) $C < 0.1$
- (b) $C > 0.1$
- (c) $C = 0.2$
- (d) None of the other options is true

Ans: (d)

Justification: The gradient descent component of weight change is $-(0.1 \times -0.01)=0.001$. The other component is $0.01c$

Now, for bypassing the local minimum we need

$$P \times 0.001 + (1-P) \times 0.01c > 0.01$$

$$\rightarrow P + 10c(1-P) > 10$$

$$\Leftrightarrow P(1-10c) > 10(1-c)$$

Since, $0 \leq P \leq 1$, this is impossible.

Q3. In an instance of BP training, a momentum factor m is used, such that the component of weight change by momentum factor is far greater than the component by gradient descent. The learning rate is 0.001 and $m=0.95$. As the number of iterations increase to a large value, the weight change value for a particular weight which was initially changed by an amount of 0.02 tends to

- (a) 0.2
- (b) 0.5
- (c) 0.4
- (d) None of given options is correct

Ans: (c)

Justification:

Since the gradient descent component of the weight change rule is insignificant compared to the component due to momentum factor, we have

$$\Delta w_{\infty} = \Delta w_0 \times (1 + m + m^2 + m^3 + \dots) = \Delta w_0 / (1 - m) = 0.02 / (1 - 0.95) = 0.02 / 0.05 = 0.4$$

Q4. In Generative AI, the tradeoff which is most specific to language models is between (choose the correct option):

- (a) Bias and variance
- (b) Precision and Recall
- (c) Capturing complex linguistic patterns and not propagating bias in training data
- (d) None of the given options is correct

Ans: (c)

Justification: (a) and (b) apply to any machine learning situation and not specifically to LLMs and GenAI. (c) indeed is characteristic of LLM and GenAI.

Q5. The property of clustering in a common representation space words from various languages that represent the same meaning is called (choose the correct option):

- (a) Isometricity

- (b) Isomorphism
- (c) Polyglossia
- (d) None of the given options is correct

Ans: (a)

Justification: by definition

Q6. Which of the following best expresses the learning rule for bias terms in a feedforward neural net (choose the correct option). η is the learning rate and δ_j is the delta value at neuron j in the network.

- (a) $\eta\delta_j^2$
- (b) $-\eta\delta_j$
- (c) $\eta.\text{sqrt}(\delta_j)$ {sqrt is the square root function}
- (d) None of the given options is correct

Ans: (b)

Justification: Apply the general weight change rule, $\Delta w_j = \eta\delta_j O_i$. The bias in a neuron is simulated by a fictitious connection from the neuron in question to another fictitious neuron whose output is fixed at -1.

Q7. The possible number of palindromes in the set of N -bit Boolean strings is (choose the correct option). $\text{Floor}(X)$ function gives the integer part of X . $\text{Ceiling}(X) = \text{floor}(X) + 1$, if X is non-integer, else X .

- (a) $2^{\text{floor}(N/2)}$
- (b) $2^{\text{ceiling}(N/2)}$
- (c) $2^{\text{sqrt}(N)}$
- (d) Cannot be determined

Ans: (a), (b)

Justification: Case-1: N is even. Take the first half of N bits' positions. These positions can be filled in $2^{N/2}$ ways. For each such string of length $N/2$, we can find a mirror image in the second half. Case-2: N odd; leave out the middle most bit, and apply the same logic for a string of length $N-1$.

Q8. Cross entropy $H(P, Q)$ between two probability distributions P and Q is not a "distance metric" in the strict sense of the term, because it violates the property of (choose the correct option):

- (a) Separability
- (b) Non-negativity
- (c) Symmetry
- (d) None of the other options is correct

Ans: (c)

Justification: Distance metrics by definition satisfy the properties of separability, symmetry, triangular equality, non-negativity etc. Cross entropy satisfies (a) and (b), but not (c)

Q9. A POS tagged corpus is given, and it is required to mark noun chunks sentencewise, i.e., sequences of determiner (*DT*), any number of adjectives (*JJ*) and a single noun (*NN*). Do not worry about multiword nouns like cricket bat. In a noun chunk only the noun is compulsory, *DT* and *JJ* are optional.

The beginning of the chunk will be labelled 1 and the rest of the words in the chunk will be labelled 0. All other words are labelled 1. Assume there are no commas separating adjectives. For example,

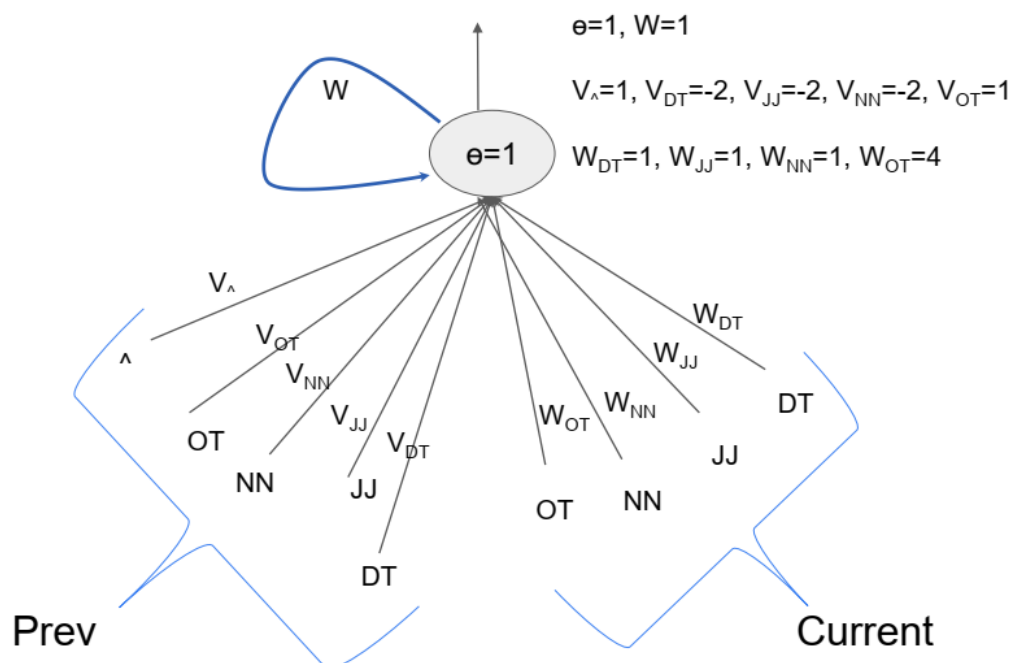
Raw corpus: *Boys play joyfully with brown balls in the green big field*

POS tagged: *Boys_NN play_VB joyfully_OT with_OT brown_JJ balls_NN in_OT the_DT green_JJ big_JJ field_NN*

Chunk: *Boys_NN_1 play_VB_1 joyfully_OT_1 with_OT_1 brown_JJ_1 balls_NN_0 in_OT_1 the_DT_1 green_JJ_0 big_JJ_0 field_NN_0*

Give a SINGLE RECURRENT PERCEPTRON to do this job, if possible. Assume there are only 4 kinds of tags: *DT NN JJ OT* (means OTHERS). *VB* is looked upon as *OT*.

Ans:



The above n/w will do the job. The perceptron gets a 4-bit 1-hot representation for the current POS and a 5-bit 1 hot representation for the previous POS. ‘^’ denotes start of a word sequence.

The following matrix gives the inequalities that must hold for the perceptron to satisfy the language constraints.

Current (W) Prev (V)	DT	JJ	NN	OT
^	$V_{\wedge} + W_{DT} > \theta$	$V_{\wedge} + W_{JJ} > \theta$	$V_{\wedge} + W_{NN} > \theta$	$V_{\wedge} + W_{OT} > \theta$
DT		$W + V_{DT} + W_{JJ} < \theta$	$W + V_{DT} + W_{NN} < \theta$	
JJ		$V_{JJ} + W_{JJ} < \theta$	$V_{JJ} + W_{NN} < \theta$	
		$W + V_{JJ} + W_{JJ} < \theta$	$W + V_{JJ} + W_{NN} < \theta$	

Current (W) Prev (V)	DT	JJ	NN	OT
NN				$V_{NN} + W_{OT} > \theta$
				$W + V_{NN} + W_{OT} > \theta$
OT	$W + V_{OT} + W_{DT} > \theta$	$W + V_{OT} + W_{JJ} > \theta$	$W + V_{OT} + W_{NN} > \theta$	$W + V_{OT} + W_{OT} > \theta$

For example, for the sentence

^ The red round ball rolled on the road

^DT JJ JJ NN OT OT DT NN

DT:

Initially state feedback: 0; Current Input: 0 0 0 1; Previous Input: 1 0 0 0 0; Net: $1.V_{\wedge} + 1.W_{DT} = 1.1 + 1.1 = 2$; $\Theta = 1$; O/P: 1

JJ:

F/B: 1; Current Input: 0 0 1 0; Previous Input: 0 0 0 0 1; Net: $1.W + 1.V_{DT} + 1.W_{JJ} = 1 - 2 + 1 = 0$; $\Theta = 1$; O/P: 0

JJ:

F/B: 0; Current Input: 0 0 1 0; Previous Input: 0 0 0 1 0; Net: $0.W + 1.V_{JJ} + 1.W_{JJ} = 0 - 2 + 1 = -1$; $\Theta = 1$; O/P: 0

NN:

F/B: 0; Current Input: 0 1 0 0; Previous Input: 0 0 0 1 0; Net: $0.W + 1.V_{JJ} + 1.W_{NN} = 0 - 2 + 1 = -1$; $\Theta = 1$; O/P: 0

OT:

F/B: 0; Current Input: 1 0 0 0; Previous Input: 0 0 1 0 0; Net: $0.W + 1.V_{NN} + 1.W_{OT} = 0 - 2 + 4 = 2$; $\Theta = 1$; O/P: 1

OT:

F/B: 1; Current Input: 1 0 0 0; Previous Input: 0 1 0 0 0; Net: $1.W + 1.V_{OT} + 1.W_{OT} = 1 + 1 + 4 = 6$; $\Theta = 1$; O/P: 1

DT:

F/B: 1; Current Input: 0 0 0 1; Previous Input: 0 1 0 0 0; Net: $1.W + 1.V_{OT} + 1.W_{DT} = 1 + 1 + 1 = 3$; $\Theta = 1$; O/P: 1

NN:

F/B: 1; Current Input: 0 1 0 0; Previous Input: 0 0 0 0 1; Net: $1.W + 1.V_{DT} + 1.W_{NN} = 1 - 2 + 1 = 0$; $\Theta = 1$; O/P: 0

Marking scheme:

Correctly working n/w: full 9 marks

Representation (1-hot or binary): 2 marks

Correct weights for the inputs and threshold: 5

Feedback weight: 2

Q10. Prove the following:

In graph search with algorithm-A (the graph is directed), open and closed lists are maintained. From the open list always that node is chosen whose f -value is the least. Now, if the heuristic function $h(.)$ satisfies triangular inequality, i.e.,

$h(\text{Parent}) \leq \text{cost}(\text{Parent}, \text{child}) + h(\text{Child})$ for every node in the search graph, then the search algorithm always terminates finding the optimal path. $\text{Cost}(\text{Parent}, \text{Child})$ means the cost of the arc between and 'Parent' - 'Child' node pair.

Recall that

$f(N) = g(N) + h(N)$, where $g(.)$ is the cost of the least cost path found so far from start node S to a node N , and $h(.)$ is the *estimate* of the distance of N from a goal node G .

Note: you can use without proof the fact that $h(N) \leq h^*(N)$ for all N , where $h^*(N)$ is the actual cost from N to G , ensures optimality

Ans: Consider a node N in the search graph. Let $N-N_1-N_2-\dots-N_k=G$ be the **optimal** path from N to G . Clearly, $h^*(N) = c(N, N_1) + c(N_1, N_2) + c(N_2, N_3) + c(N_3, N_4) + \dots + c(N_{k-1}, G)$. $c(.)$ is the cost.

Now,

$$h(N) \leq h(N_1) + c(N, N_1)$$

$$h(N_1) \leq h(N_2) + c(N_1, N_2)$$

$$h(N_2) \leq h(N_3) + c(N_2, N_3)$$

...

$$h(N_{k-2}) \leq h(N_{k-1}) + c(N_{k-2}, N_{k-1})$$

$$h(N_{k-1}) \leq h(G) + c(N_{k-1}, G); h(G) = 0$$

Adding these inequalities, $h(N) \leq c(N, N_1) + c(N_1, N_2) + c(N_2, N_3) + c(N_3, N_4) + \dots + c(N_{k-1}, G)$

Hence proved.

Marking scheme:

5 marks for faultless proof, else 0.

=====Paper Ends=====